JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 94, NO. B5, PAGES 6075–6084, MAY 10, 1989

Paleomagnetic and Cooling Constraints on the Duration of the Pahute Mesa-Trail Ridge Eruptive Event and Associated Magmatic Evolution, Black Mountain Volcanic Center, Southwestern Nevada

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Two successive ash-flow cooling units of the Black Mountain volcanic center, the Pahute Mesa and Trail Ridge members of the late Miocene Thirsty Canyon Tuff, possess nearly indistinguishable reverse polarity remanent magnetization directions of about 178° declination and -10° inclination. These are interpreted to reflect thermoremanent magnetization either during a single geomagnetic field excursion or polarity transition, or possibly from a field oriented near the limit of secular variation. Regardless of the type of geomagnetic field behavior, the Pahute Mesa Member was apparently overlain by the Trail Ridge Member before appreciable shift of the anomalous geomagnetic field. This strongly suggests that the time between eruption of the two ash-flow sheets was very short, perhaps less than 100 years, a conclusion that is independently supported by the local absence of a complete cooling break between the two units. The interval between eruption of the Pahute Mesa and Trail Ridge members was too brief for reestablishment of chemical stratification in the Black Mountain magma system. Nevertheless, a small volume of highly evolved comendite magma developed and was erupted as lava within the Black Mountain caldera and as a component of the Trail Ridge Member. The short repose interval and rapid development of the comendite, together with evidence for mixing of magmas of silicic to near-mafic composition (Vogel et al., this issue), and the presence of air-fall tuff at the base of the Trail Ridge, lead us to conclude that eruption of the Trail Ridge Member was probably triggered by injection of hot, relatively unevolved magma into the Black Mountain magma body.

INTRODUCTION

As the result of its youth, geological and geochemical complexity, and excellent state of preservation and exposure, the late Miocene Black Mountain volcanic center in southern Nevada [*Noble et al.*, 1984] (Figure 1) is well suited for the study of the evolution of a high-level magma system [e.g., *Vogel et al.*, 1987, this issue]. The rock units and history of the Black Mountain center as presently recognized are summarized in Table 1.

In this paper we focus on the second and third of the four ash-flow sheets of the Black Mountain center, the Pahute Mesa and Trail Ridge members of the Thirsty Canyon Tuff. These two units are identical in age within the limits of uncertainty of the potassium-argon method (Table 2) and are, in nearly all localities, separated by an unambiguous complete cooling break. The ash-flow sheets are lithologically very similar, and chemical and mineralogical data are consistent with eruption of both sheets from the same complexly zoned magma body [*Vogel et al.*, 1983; 1987; this issue]. In addition, a small but significant body of highly evolved comendite magma developed after eruption of the Pahute Mesa Member, and was erupted as intracaldera lava and as a component of the Trail Ridge Member. This sequence of temporally constrained magmatic and volcanic events, beginning with eruption of the Pahute Mesa Member and concluding with eruption of the Trail Ridge Member, can be informally considered as comprising a genetically meaningful "Pahute Mesa-Trail Ridge eruptive event".

Paleomagnetic data, initially acquired to corroborate stratigraphic and source-area reinterpretations based on field relations and geochemical fingerprinting [*Noble et al.*, 1983; 1984], provide important evidence for the very small amount of time between eruption of the Pahute Mesa and Trail Ridge Members, possibly on the order of one hundred years [*Weiss and Noble*, 1983]. Other volcanic centers that erupted mildly to moderately peralkaline rhyolitic rocks and appear to have undergone periods of rapid magma evolution include the Sierra La Pimavera, Mexico [*Mahood and Drake*, 1982], the Menengai system in Kenya [*Leat et al.*, 1984] and the Kane Springs Wash [*Novak*, 1984] and Stonewall Mountain [*Noble et al.*, 1988; *Weiss and Noble*, this issue] volcanic centers in Nevada. Such a small repose interval implies extremely rapid rates and mechanisms of magma evolution (differentiation, zoning, mixing, etc.) [e.g., *Vogel et al.*, 1983; 1987; this issue], and bears directly on questions concerning the mechanism(s) that trigger large-volume pyroclastic eruptions.

STRATIGRAPHIC, COOLING AND CHEMICAL RELATIONS

Individual ash-flow sheets are ideal time-stratigraphic units because they are the products of single, catastrophic pyroclastic eruptions, and thus are deposited virtually simultaneously over broad areas. Where unaffected by movement after aquisition of thermoremanent magnetization (TRM), or by later remag-

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Paper number 89JB00006.
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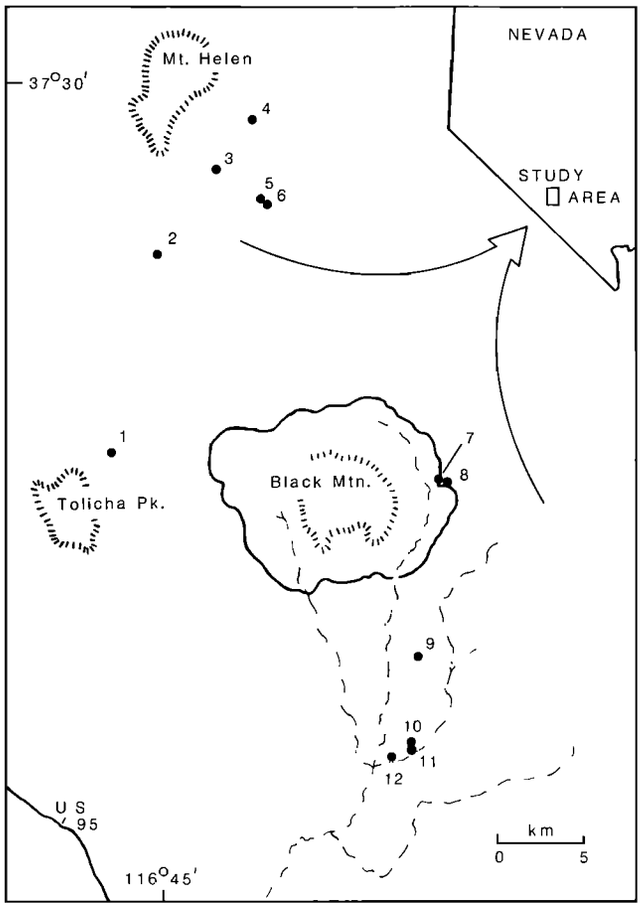


Fig. 1. Map showing the location of the Black Mountain volcanic center. Numbered dots show the location of paleomagnetic sample sites.

TABLE 1. Summary of the Stratigraphy and Caldera Evolution of the Black Mountain Volcanic Center, Southern Nevada

Stratigraphic Unit	Description
Trachyte of Hidden Cliff	trachyte and mafic trachyte lavas; age >6.3 ± 0.2 Ma
Gold Flat Member of the Thirsty Canyon Tuff; caldera collapse	pantellerite ash-flow tuff
Lavas of Pillar Spring	trachyte, comendite, and subalkaline rhyolite lavas
Rocks of Yellow Cleft; possible central resurgence	trachyte and comendite lavas; syenite
Trail Ridge Member of the Thirsty Canyon Tuff; probable renewed caldera collapse	comendite ash-flow tuff; K-Ar age of 7.5 ± 0.2 Ma
Pahute Mesa Member of the Thirsty Canyon Tuff; caldera collapse	subalkaline rhyolite ash-flow tuff; K-Ar age of 7.8 ± 0.2 Ma
Rocket Wash Member of the Thirsty Canyon Tuff; probable caldera collapse	subalkaline rhyolite ash-flow tuff; K-Ar age of 7.8 Ma
Lavas of Ribbon Cliff	trachyte to rhyolite lavas

Modified from *Noble and Christiansen* [1974], *Noble and Parker* [1975], *Noble et al.* [1984], and *Noble and Weiss* [1986]. Ages of the Trail Ridge and Pahute Mesa members are based on new K-Ar age determinations (Table 2). Age of the Rocket Wash Member is the average of the age of 7.7 Ma (recalculated using current constants) reported by *Kistler* [1968] and an age of 8.0 Ma obtained more recently on the same sanidine concentrate [*Carr*, 1984].

netization, ash-flow cooling units provide spot recordings (TRM directions) of the geomagnetic field orientation at the time of deposition [e.g., *Gose*, 1970; *Grommé et al.*, 1972; *Reynolds*, 1977; *Geissman*, 1980; *Reynolds et al.*, 1986; *Rosenbaum*, 1986]. Preserved TRM directions have thus been used by many workers as a tool to aid in the definition and identification of individual ash-flow sheets and in their correlation over wide areas [e.g., *Noble et al.*, 1968; *Grommé et al.*, 1972; *Nairn et al.*, 1975; *Shuey et al.*, 1976; *Gillett*, 1982]. TRM directions have also been used to provide evidence on the duration of pyroclastic eruption and timing of caldera resurgence [*Reynolds et al.*, 1986]. Magnetic intensities are commonly high [e.g., *Bath*, 1968], and direct field measurement of magnetic polarity is routinely used as a tool for field identification and correlation of ash-flow sheets.

The tuffs sampled for this study have undergone no discernable tectonic rotation, and thus no structural corrections have been applied. The cooling units at the sample sites vary from a few meters to about 80 m thick; evidence of rheomorphism has not been observed. The nature of the units sampled as well as that of stratigraphically higher units that previously overlay them in several localities, in conjunction with the vertical position of the sample localities within the units, preclude reheating of the sampled tuffs above the range of remanence blocking temperatures. None of the specimens exhibit any megascopic or microscopic evidence of hydrothermal alteration. Hematite in devitrified and(or) vapor-phase crystallized specimens is primary, reflecting oxidation as the result of the loss of hydrogen gas during primary, high-temperature crystallization [*Noble et al.*, 1984; *Schlinger et al.*, 1988]. This primary hematite is not of "authigenic" origin, in contrast to that found by *Reynolds* [1977] in certain tuffs of the Yellowstone Group. The glassy rocks have, as have virtually all phenocryst-bearing, glassy, silicic rocks of Tertiary age, been

TABLE 2. New Potassium-Argon Age Determinations on the Pahute Mesa and Trail Ridge Members of The Thirsty Canyon Tuff

Specimen	K_2O, wt %	$^{40}Ar^*$ kmol/kg x 10^{-11}	$^{40}Ar^*$ %	Age ±1σ Ma
1†	7.23	7.8358	38.3	7.5 ± 0.2
2‡	7.22	8.1674	80.1	7.8 ± 0.2

Analyses made in laboratories of the U.S. Geological Survey, Menlo Park, California, under the supervision of E. H. McKee.

Constants: $\lambda_\epsilon = 0.581 \times 10^{-10}\ yr^{-1}$, $\lambda_\beta = 4.962 \times 10^{-10}\ yr^{-1}$, $^{40}K/K_{(total)} = 1.167 \times 10^{-4}$ kmol/kmol.

†Sanidine phenocryst separate from large, slightly vesiculated glassy block of cognate magmatic material (latitude 37°8.5'N, longitude 116°33.8'W) of the Trail Ridge Member of the Thirsty Canyon Tuff. Field number Ttt-BLOB.

‡Sanidine phenocryst separate from large, slightly vesiculated glassy block of cognate magmatic material (latitude 37°8.4'N, longitude 116°33.8'W) of the Pahute Mesa Member of the Thirsty Canyon Tuff. Field number Tts-BLOB.

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hydrated as a result of exposure to atmospheric moisture [*Noble*, 1968].

A well-defined, complete, and readily identifiable cooling break [*Smith*, 1960; *Noble et al.*, 1968] is present at almost all localities between the largely primarily devitrified and vapor-phase crystallized Pahute Mesa and Trail Ridge members [*Noble et al.*, 1964]. This complete cooling break is demonstrated by the presence of glassy air-fall and nonwelded to partly welded ash-flow tuff, composing the lowermost part of the Trail Ridge Member and, in a few places, by slightly welded glassy ash-flow tuff in the uppermost part of the Pahute Mesa Member. At some localities in Thirsty Canyon between latitudes 37°05.5' and 37°07'N, however, where both members thicken within a preexisting depression probably related to earlier, pre-Black Mountain caldera collapse [*Byers et al.*, 1976; D. C. Noble, unpublished data, 1965], the complete cooling break is absent [*Noble and Weiss*, 1986]. The porous ash-flow and, locally, air-fall tuff that is elsewhere glassy have been completely and pervasively affected by vapor-phase crystallization, resulting in a partial, rather than a complete, cooling break. In a few localities, vapor-phase crystallization was not complete; the finer-grained groundmass material was crystallized, but the pumice fragments remained glassy. The unusually thick accumulation of tuff of the Pahute Mesa Member in this area remained hot and was still releasing sufficient vapor from primary crystallization of supercooled melt to cause vapor-phase crystallization of the porous glassy tuff at the contact of the two units. The Pahute Mesa and Trail Ridge members thus comprise a composite sheet as defined by *Smith* [1960].

Chemical studies of glassy pumice fragments by *Vogel et al.* [1987, this issue] have shown that the ash-flow tuffs of the Trail Ridge Member were derived mostly from the same, chemically less evolved layer of magma that had been in part erupted to form the upper part of the Pahute Mesa Member. Apparently, the continuity of chemical stratification in the Pahute Mesa magma body was not reestablished prior to eruption of the Trail Ridge Member.

A chemically distinct magma did, however, develop in the time between eruption of the Pahute Mesa and Trail Ridge members. In many localities the Trail Ridge Member includes at its base a subunit of air-fall tuff that locally reaches a thickness of several meters or more. In places this subunit includes pumice fragments, which based on color and direct chemical analysis are of different compositions, and contains common to abundant lithic fragments. Many of these lithic fragments consist of obsidian, in some cases nonhydrated, of highly evolved peralkaline rhyolite (comendite) composition [cf. *Noble et al.*, 1979; *Drexler et al.*, 1983] and of granophyrically crystallized comendite lava of similar original composition containing abundant groundmass sodic amphibole and pyroxene (Table 3, specimens 1 and 2). Similar lithic fragments are not present in the Pahute Mesa Member. These fragments were therefore most probably derived from domes of comendite lava emplaced within the Black Mountain caldera after eruption of the Pahute Mesa Member. Some of the glassy blocks of pumice in the air-fall unit at the base of the Trail Ridge Member and in pyroclastic flows probably assignable to the Trail Ridge Member (Table 3, specimens 3 and 4) have major and minor element compositions virtually identical to those of the comendite obsidian fragments. Although these hydrated glassy specimens have suffered changes in alkali and alkaline earth contents after cooling as the result of interaction with groundwater [e.g, *Jezek and Noble*, 1978, and references cited therein], the low Al_2O_3 and high Fe, Ti, Zr, and heavy rare earth element contents of the specimens, in combination with the close similarity in these and other elements to those of the nonhydrated comendite glasses, strongly suggest an original peralkaline chemistry. Further evidence for peralkaline chemistry within the Trail Ridge Member is provided by the presence of groundmass sodic amphibole and pyroxene within certain primarily crystallized welded tuffs of the member [*Noble et al.*, 1984]. Evidence for peralkaline chemistry has not been recognized in rocks of the Pahute Mesa Member; it therefore appears that comendite magma developed within the Black Mountain magma chamber, without reestablishment of the earlier chemical stratification, during the time between eruption of the Pahute Mesa and Trail Ridge members.

PALEOMAGNETIC PROCEDURES

Field Sampling

At seven of the twelve sample sites (Figure 1) core samples were drilled and oriented in situ using a Brunton compass and drill hole clinometer. Block samples oriented by Brunton compass were collected at sites 2, 5, 6, and 9 and drilled in the laboratory. Samples from sites 2, 3, 4, 7, 10, 11, and 12 consist of densely to partially welded, completely crystallized ash-flow tuff. These samples have all undergone primary devitrification during the original cooling of the ash-flow units. The samples from sites 2 and 11 are moderately welded and show evidence of vapor-phase crystallization at elevated temperatures in addition to primary devitrification. Samples from site 8 are from a near-vent, welded and devitrified air-fall facies of the Trail Ridge Member. Samples from sites 1 and 9 consist of moderately welded and densely welded glassy tuff, respectively.

Samples from sites 5 and 6 are large blocks of partly vesiculated glassy pumice from a unit of nonwelded to partly welded ash-flow tuff, the Dry Lake Member of *Rogers et al.* [1968]. This unit may belong either to the Pahute Mesa Member or to the Trail Ridge Member, or may possibly be a distinct cooling unit younger than the Pahute Mesa Member. Although the precise stratigraphic position of the Dry Lake Member is not known [*Noble et al.*, 1984], data from these specimens are included because they provide additional evidence for the occurrence of an anomalous orientation of the geomagnetic field during development of the Black Mountain center.

Laboratory Measurements

Natural remanent magnetization directions were measured with a computerized Schonstedt fluxgate magnetometer. Magnetic cleaning by progressive alternating field (AF) demagnetization procedures, with peak fields of as much as 98 mT, was used to isolate inferred thermoremanent magnetization (TRM) directions from the effects of viscous remanent magnetization. Representative specimens from each unit were subjected to progressive thermal demagnetization to verify the directions identified by AF demagnetization, to test remanence stability, and to attempt to identify the dominant magnetic carrier phases.

Comparison of thermal and AF demagnetization curves (Figure 2) with those obtained from tuffs by other workers, who also performed hysteresis and polished section petrographic analyses, suggests that magnetite, hematite, and titanium-rich hematite(?) [cf. *Rosenbaum*, 1986] are the dominant magnetic carrier phases (C. S. Grommé, personal communication, 1987). The response of specimens to progressive demagnetization (Figure 3) and the primary origin of the hematitic material is in-

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TABLE 3. Major Element and Selected Minor Element Analyses of Comendite Lithic and Pumice Fragments From the Trail Ridge Member of the Thirsty Canyon Tuff

	Sample			
	1	2	3	4
SiO_2	71.20	74.17	72.2	72.4
Al_2O_3	12.45	12.27	12.0	12.8
Fe_2O_3	1.35	1.90	3.38*	3.65*
FeO	2.26	1.17	--	--
MgO	0.01	0.04	0.24	0.56
CaO	0.36	0.10	0.50	0.42
Na_2O	5.66	4.88	4.50	4.21
K_2O	4.70	4.55	6.0	5.7
H_2O^+	0.29	0.15	--	--
H_2O^-	0.00	0.06	--	--
TiO_2	0.22	0.17	0.22	0.23
MnO	0.14	0.09	0.15	0.11
P_2O_5	--	0.02	--	--
Cl	0.25	0.01	--	--
F	0.40	0.16	--	--
Subtotal	99.29	99.74	99.2	100.1
Less O	0.22	0.07	--	--
Total	99.07	99.67	99.2	100.1
Ba	3[a]	30[a]	1[b]	19[b]
Be	8[a]	10[a]	11[c]	11[c]
Cr	2[a]	2[a]	1[b]	1[b]
La	208[d]	160[a]	181[d]	160[d]
Nb	160[a,e]	140[a,e]	86[b]	96[b]
Rb	283[d]	--	284[d]	284[d]
Sc	1.4[d]	3[a]	2.4[d]	1.1[d]
Sr	<2[a]	10[a]	1[b]	5[b]
Th	41.4[d]	41.6[d]	40.2[d]	41.3[d]
Yb	14.2[d]	9[a]	12.5[d]	12.7[d]
Zr	1400[a,e]	1400[a,e]	1030[b]	1070[b]
Eu/Eu*	0.06	0.03	0.05	0.10

Sample 1, lithic fragments of nonhydrated comendite glass in coarse air-fall tuff at base of the Trail Ridge Member east of Black Mountain. Major element values are averages of classic wet-chemical analyses of specimens Ttt-ob-g and Ttt-ob-b, E. L. Munson, U.S. Geological Survey, analyst. Al_2O_3 is not corrected for Zr, REE, Nb, etc., oxides. Instrumental neutron activation analyses (INAA) were obtained on specimen U20K-TO. Sample 2, lithic fragment of granophyrically crystallized comendite lava (BM-GV) containing groundmass sodic amphibole and pyroxene in coarse air-fall tuff at base of Trail Ridge Member directly north of the Black Mountain caldera. Major element analysis by classic wet-chemical methods, G. O. Riddle, U.S. Geological Survey, analyst. Al_2O_3 is not corrected for Zr, REE, Nb, etc., oxides. Note that Na_2O has been lost from this specimen as from most crystallized peralkaline rhyolites [*Noble*, 1970]. Sample 3, large blocks of comendite pumice from the "Dry Lake Member" north of Black Mountain. Major-element values are averages of analyses obtained on specimens S-17-1 and S-17-5 by Barringer Laboratories, Inc., using induction-coupled plasma (ICP) methods. Sample 4, composite sample of dark comendite pumice (TR-2-1) from coarse air-fall tuff at base of the Trail Ridge Member south of Black Mountain. Major element analysis by Barringer Laboratories using ICP methods.

* Total iron as Fe_2O_3

[a] Analyses by quantitative optical emission spectrographic methods, J. Haffty and H. Neiman, U.S. Geological Survey, analysts.

[b] Analyses by X-ray fluorescence (XRF) methods, T. A. Vogel, analyst.

[c] Analyses by Barringer Laboratories, Inc., using ICP methods.

[d] Analyses carried out at the Lawrence Livermore National Laboratory using INAA methods [*Vogel et al.*, 1983].

[e] Values for Nb and Zr obtained on other specimens by XRF methods are about 30-35% higher than values previously obtained on the same specimens by J. Haffty and H. Neiman using optical emission spectrographic methods, and thus these values should be reduced by about 25% for comparision with XRF values given for samples 3 and 4.

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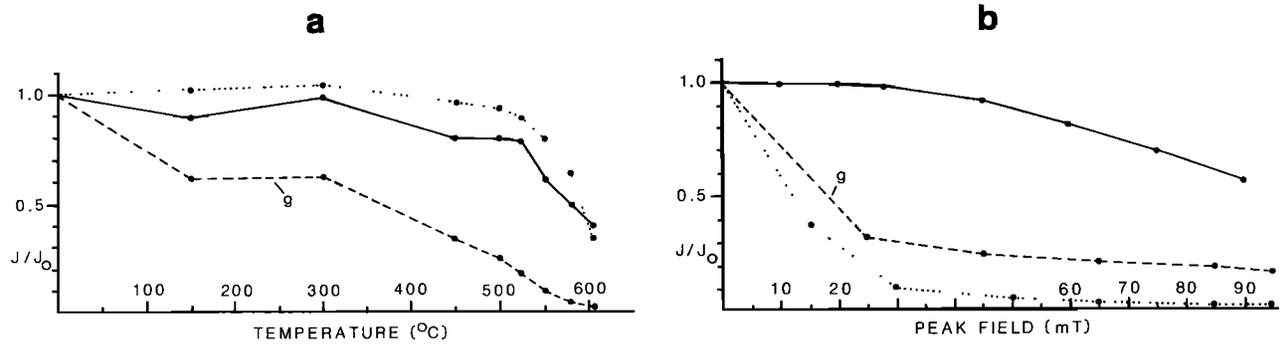


Fig. 2. (*a*) Alternating field and (*b*) thermal demagnetization curves for representative specimens of the Pahute Mesa (dotted lines), Trail Ridge (solid lines) and Dry Lake(?) (dashed lines) members. J/J_o is normalized magnetic intensity; g is glassy groundmass.

terpreted to be consistent with the existence of a stable TRM, carried by both magnetite and hematite, acquired during the cooling of the ash-flow units. Six core specimens that failed to reach stationary, unchanging magnetization directions under progressive AF or thermal demagnetization were considered magnetically unstable and were not included in subsequent calculations.

The stable TRM directions measured for each specimen were averaged to obtain calculated site mean directions (Table 4 and Figure 4). Unit mean TRM directions (Table 5 and Figure 5) were calculated both by averaging the calculated site mean TRMs and by averaging the individual specimen TRMs from each unit.

INTERPRETATION OF PALEOMAGNETIC DATA

The response to both AF and thermal demagnetization, the clustering of resultant magnetization directions of specimens from different zones of the ash-flow units at localities tens of kilometers apart, the primary origin of both magnetite and groundmass hematite, and the lack of megascopic or microscopic evidence of alteration or deformation together strongly suggest that the measured directions are indeed stable thermoremanent magnetization directions acquired upon cooling in the prevailing geomagnetic field shortly after emplacement of each of the two ash-flow sheets. Densely welded glassy portions of ash-flow units have been considered by some

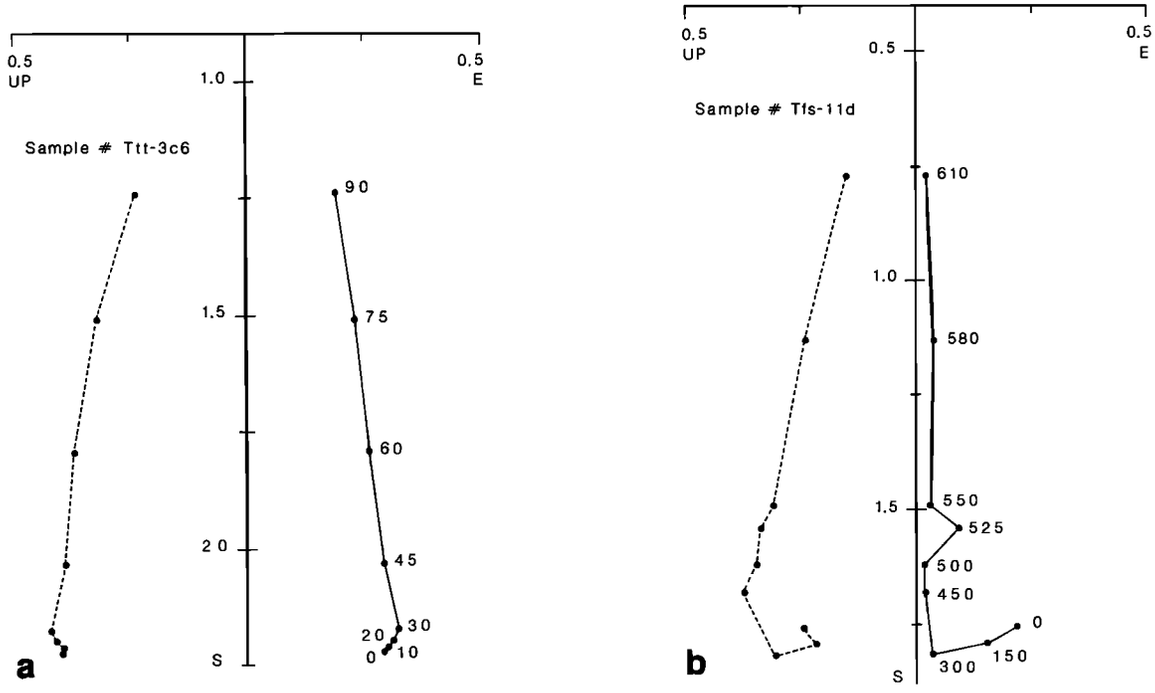


Fig. 3. Representative vector plots showing response to (*a*) alternating field and (*b*) thermal demagnetization. Solid and dashed lines indicate horizontal and vertical vectors, respectively; numbers show peak field (mT) and temperature (°C) applied at each step of demagnetization. Axes are $x10^{-5}$ A m^2 for Figure 3*a* and $x10^{-7}$ A m^2 for Figure 3*b*.

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TABLE 4. Site Mean TRM Directions

Site	Location		N/No	J/Jo	Declination, deg	Inclination, deg	α_{95}	K
	Latitude °N	Longitude °W						
				Pahute Mesa Member				
2	37°24.8'	116°45.3'	6/6	0.02	161.4	-2.4	15.9	18
3	37°27.6'	116°42.9'	1/2	0.20	178.9	-10.2	--	--
4	37°29.2'	116°41.5'	7/9	0.02	187.1	-1.2	10.5	34
7	37°17.8'	116°34.4'	5/5	0.26	176.9	-11.1	14.4	29
11	37°09.6'	116°35.6'	5/5	0.14	188.1	-15.7	12.7	37
				Trail Ridge Member				
1†	37°18.7'	116°47.0'	4/4	0.15	172.8	3.1	20.6	21
8	37°17.7'	116°34.1'	1/3	0.54	167.5	-22.0	--	--
9†	37°12.5'	116°35.2'	7/7	0.06	180.6	-11.4	8.3	54
10	37°09.7'	116°35.5'	4/5	0.13	179.4	-23.3	13.5	47
12	37°09.3'	116°36.2'	7/7	0.16	178.1	-15.9	7.6	65
				Dry Lake Member(?)				
5‡	37°26.7'	116°41.1'	5/5	0.43	169.5	-15.4	7.9	94
6‡	37°26.6'	116°41.0'	5/5	0.02	176.1	-9.6	9.1	71

N, the number of specimen TRMs used in the calculation; J/J_o, the lowest normalized magnetic intensity achieved by demagnetization at each sample site; declination is in degrees clockwise from north; inclination is in degrees from the horizontal, negative sign indicates reverse magnetic polarity; α_{95}, the semiangle defining the cone of 95% confidence; K, precision parameter of *Fisher* [1953].

† Vitrophyre.

‡ Glassy nonwelded tuff.

workers [e.g., *Reynolds*, 1977; *Geissman*, 1980] to yield the most reliable and precise TRM directions. However, as shown in Table 4, Table 6, and Figure 4, dispersion in TRM directions obtained from vitrophyre is similar to the dispersion observed in devitrified portions of the cooling unit. Such dispersion, commonly observed in ash-flow cooling units [e.g., *Gose*, 1970; *Nairn et al.*, 1975; *Reynolds*, 1977; *Geissman*, 1980], is attributable to some combination of post-blocking-temperature grain rotation, orientation errors, and(or) measurement errors. In view of the uncertainties introduced during collection and measurement, and the possibility of some post-blocking-temperature grain rotation because of the abundant primary high-blocking-temperature hematitic material present in some

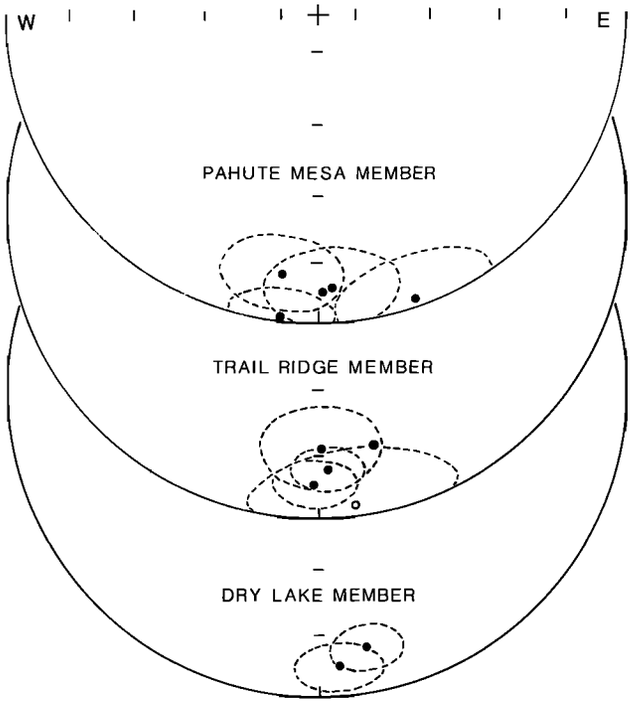


Fig. 4. Equal-area projection of calculated site mean TRM directions for Black Mountain units. Solid circles indicate upper hemisphere projection (reverse polarity) and open circle indicates lower hemisphere projection (normal polarity). Dashed lines show projections of cones of 95% confidence.

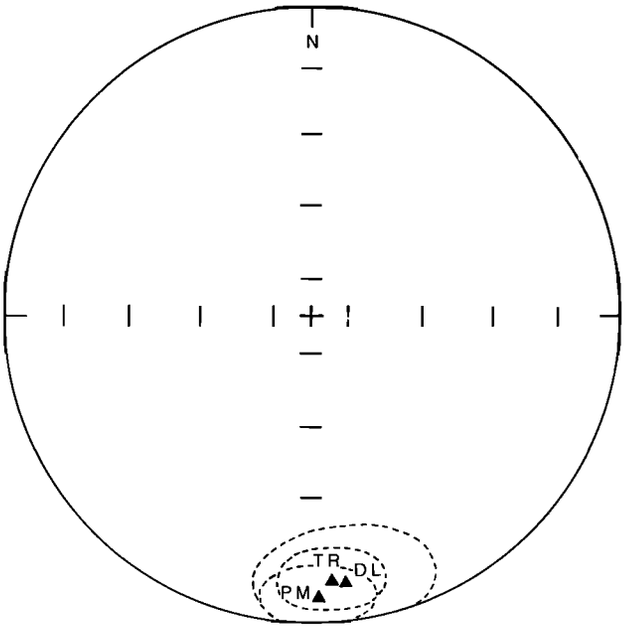


Fig. 5. Equal-area upper hemisphere projection (reverse polarity) of unit mean TRM directions. PM, TR and DL indicate Pahute Mesa, Trail Ridge and Dry Lake(?) members, respectively. Dashed lines show projections of the cones of 95% confidence intervals.

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TABLE 5. Unit Mean TRM Directions and Calculated VGP Positions

Based on Site Mean TRMs					Based on Total Stable Specimens					VGP Position	
N	Declination, deg	Inclination, deg	α_{95}	K	N	Declination, deg	Inclination, deg	α_{95}	K	Latitude °S	Longitude °W
					Pahute Mesa Member						
5	178.4	-8.2	11.7	44	24	178.5	-7.1	7.0	19	56.8°	113.7°
					Trail Ridge Member						
5	175.6	-14.0	11.3	47	23	177.7	-12.9	5.4	32	59.6°	107.9°
					Dry Lake Member (?)						
2	172.7	-12.5	19.0	175	10	172.8	-12.5	5.8	71	58.4°	102.7°

α_{95}, and K expressed the same as in Table 4; N, the number of site means or specimen TRMs used to calculate each unit mean TRM; VGPs calculated from unit mean TRMs derived from site mean TRMs.

specimens (Figure 2), the observed dispersion does not require temporal variations in the geomagnetic field during the times of ash-flow sheet deposition. It is unlikely that significant temporal variation occurred in the small time interval, probably on the order of 10 to 10^2 years, necessary for cooling all but the thickest sections of these thin ash-flow sheets below remanence blocking temperatures [cf. *Riehle*, 1973].

It has been suggested that the observed shallow TRM inclinations may be due to flattening of an originally high-angle inclination during compaction and welding of the tuffs. *Rosenbaum* [1986] related systematic variation of remanence directions with depth, in a welded portion of the Topopah Spring Member of the Paintbrush Tuff, to compaction and welding of thick (300 m) sections of tuff, in part at temperatures inferred to be lower than the blocking temperatures of a hematitic magnetic carrier phase. Such rotation is unlikely in the cooling units sampled for our study because they are only a few meters to a few tens of meters thick. Additionally, the degree of compaction and welding of the Pahute Mesa and Trail Ridge members varies from moderate (sample sites 1, 2, 3, 10, and 12) to dense (sample sites 7, 8, and 9), without an associated flattening of the inclinations (Table 4). In particular, the blocks of glassy, noncompacted pumice randomly oriented within the nonwelded to partly welded matrix of the Dry Lake Member (sample sites 5 and 6, Table 4) yield low-angle inclinations identical within the limits of experimental uncertainty to those of the welded specimens. Other workers [e.g., *Reynolds*, 1977; *Gillett*, 1982; *Weiss and Noble*, 1983, this issue] have found that preserved TRM directions in densely welded zones of other silicic, subalkaline and peralkaline ash-flow sheets may have consistently near-expected, shallow, or very shallow inclinations, of either normal or reverse polarity, relative to expected directions for the studied areas. Such observations, in addition to results from the Tiva Canyon Member of the Paintbrush Tuff [*Rosenbaum*, 1986], provide evidence to suggest that TRM acquisition in some ash-flow units occurs largely or entirely after compaction and welding and therefore that the TRM directions have not been flattened.

TABLE 6. Variation in TRM Directions From Sample Sites 9 and 10

Specimen	H	J/J_o	Declination, deg	Inclination, deg
		Site 9 (Vitrophyre)		
F83Ttt3c1	80	0.66	190.2	-11.4
3c2	65	0.78	192.4	-11.5
3c3	70	0.74	171.8	-11.4
3c5	61	0.39	172.1	-12.3
3c6	90	0.56	171.2	-10.7
3c11	85	0.65	173.1	- 9.7
3c12*	600°C	0.06	188.2	-12.3
		Site 10 (Welded, Complete Primary Crystallization)		
Ttt-12a1†	35	0.57	184.1	-31.9
12a2†*	625°C	0.13	175.3	-11.6
12b	32.5	0.63	178.1	-33.9
12d	35	0.57	181.0	-15.7

H, peak alternating field (mT) or temperature (°C) applied; J/J_o, normalized magnetic intensity after demagnetization.

* Thermal demagnetization.

† Specimens cut from the same core.

Rosenbaum [1986] also noted the negligible influence of flow and compaction-related anisotropy of magnetic susceptibility [e.g., *Ellwood*, 1982; *Knight et al.*, 1986] on TRM directions. The similar directions of remanence found in nonwelded, moderately welded, densely welded, and both glassy and crystallized portions of the Pahute Mesa, Trail Ridge, and Dry Lake(?) members and the small thicknesses of the units strongly suggest that the TRM directions were acquired largely or entirely after completion of compaction and welding and therefore do not reflect flattening of prewelding TRM directions.

Regional and local geologic relations indicate negligible post-8 Ma translation or tilting of the study area. The calculated unit mean TRM directions are therefore considered to provide an accurate record of an anomalous, near-horizontal geomagnetic field orientation in southwestern Nevada during the cooling of these units. The results obtained for the Pahute Mesa Member are identical within the limits of experimental uncertainty to those obtained by *Noble et al.* [1968].

Assuming an axial geocentric dipole field model for the long-term behavior of the geomagnetic field, virtual geomagnetic poles (VGPs) located near 113° west longitude, 57° south latitude (Table 5) would produce the observed unit mean TRM directions. These VGP locations are about 30° in both latitude and longitude from those expected for a reverse-polarity, late Miocene pole position [*Irving and Irving*, 1982]. Similar anomalous low-angle TRM directions have been related by some workers to intermediate field orientations during geomagnetic excursions and polarity transitions [e.g., *Reynolds*, 1977; *Mankinen et al.*, 1985; *Hoffman*, 1986]. Other workers,

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particularly those concerned with tectonic studies or the long-term behavior of the geomagnetic field, consider only VGPs more than 40°-45° from the expected mean pole position to represent geomagnetic excursions or polarity transitions [*Merrill and McElhinny*, 1983]. By this convention the low-angle TRMs and corresponding VGPs observed in this study are the result of extreme secular variation. Additionally, the observed TRM directions may not be confidently attributed to an excursion or polarity transition because data are not currently available to bracket the Pahute Mesa, Trail Ridge, and Dry Lake members between stable periods of expected field orientation of either polarity, showing completion of an excursion or transition and excluding secular variation as the cause of the anomalous field orientation. In this case, further resolution of the problem is unnecessary because excursions, transitions, and secular variation when the field is far from the mean position are all known for rapid changes in field orientation; thus they all allow resolution of geologically short intervals of time and enable us to place constraints on the timing of eruptive activity at Black Mountain.

Both rapid directional changes of as much as 58° ± 21° per year, and periods of directional stability and recurring intermediate directions have been identified in detailed records of complete transitions and excursions [*Mankinen et al.*, 1985; *Hoffman*, 1986, 1988; *Laj et al.*, 1988]. In spite of the great variability of field behavior within and among transitions and excursions, "best guess" estimates of the duration of a complete transition are on the order of 1,000-10,000 years, depending in part on the criteria (anomalous decrease in total field intensity or anomalous pole position) used to define the end points of the transition [e.g., *Merrill and McElhinny*, 1983; *Clement and Kent*, 1987]. More detailed discussions of geomagnetic field behavior during excursions and polarity reversals are given by *Hoffman* [1988] and *Bogue and Hoffman* [1988].

Changes in the geomagnetic field as the result of secular variation are better documented than those from transitions or excursions. When the geomagnetic field is far from the expected time-averaged mean position, changes in field orientation are commonly on the order of 10° or more per century [*McElhinny*, 1973; *McElhinny and Merrill*, 1975; D. E. Champion, personal communication, 1987].

CONCLUSIONS

Time Between Eruption of Pahute Mesa and Trail Ridge Members

The nearly identical unit mean TRM directions measured for the Pahute Mesa and Trail Ridge members of the Thirsty Canyon Tuff are unlikely to record more than one episode of anomalous field orientation. It appears that the two cooling units were emplaced either during a period of extreme secular variation or within a transition or excursion. Irregardless of the type of geomagnetic field behavior, the short time frame discussed above, the rapid rates of directional changes, and the very small differences in unit mean TRM directions (and VGPs) allowed by the data clearly imply an extremely short time between emplacement of these ash-flow sheets. Although long enough to develop a complete cooling break in nearly all localities, this interval appears to have been too brief to have allowed a significant change in the orientation of the anomalous geomagnetic field. Therefore, based on the very small difference in unit mean TRM directions and the rapid rates at which the geomagnetic field orientation changes when located far from the time-averaged mean position, we conclude that the time between deposition of the Pahute Mesa and Trail Ridge members was less than about 10^4 years and perhaps was on the order of 100 years or less. The local absence of a complete cooling break between the Pahute Mesa and Trail Ridge members independently supports the above conclusion.

Constraints on Magma Evolution and Eruptive Mechanisms

The very short time between eruption of the Pahute Mesa and Trail Ridge members indicates a very rapid development of the peralkaline magmas and limits possible mechanisms for initiating eruption of the Trail Ridge Member. The peralkaline magmas appear to have composed a subordinate part of the magma that erupted to form the Trail Ridge Member, much of which consisted of relatively unevolved magma similar to that which formed the upper part of the Pahute Mesa Member. Indeed, within the lower part of the Trail Ridge, the comendite pumice fragments are accompanied, in both the ash-flow and air-fall units, by fragments of pumice that are chemically typical of the bulk of the Pahute Mesa and Trail Ridge members, showing that both magma types erupted simultaneously. Large-scale chemical stratification was therefore not reestablished, and it is unlikely that an unstable, volatile-rich cap accumulated at the top of the magma body that was sufficiently large to trigger the eruption of the Trail Ridge Member [cf. *Smith*, 1979].

The comendite magma that developed after eruption of the Pahute Mesa Member was the first silicic magma of peralkaline composition to develop at the Black Mountain center. Peralkaline chemical character became progressively more pronounced during the later stages of activity of the Black Mountain center; the Gold Flat Member is of highly peralkaline pantelleritic composition [*Noble*, 1965], and intermediate and silicic rocks of the Lavas of Pillar Spring and the Lavas of Hidden Cliff in many cases contain sodic amphibole and pyroxene in the groundmass.

The magmas of peralkaline chemical character and the presence of an appreciable volume of air-fall tuff at the base of the Trail Ridge Member, together suggest that both the generation of peralkaline magma and the eruption of the Trail Ridge Member were triggered by rapid, large-scale events within the Black Mountain system. *Vogel et al.* [1987, this issue] have presented persuasive evidence for major and repeated mixing of magma, some of which was of low-silica intermediate to mafic composition, at the Black Mountain center. The availability of mafic and low-silica intermediate magma is also documented by the local presence of lava flows of hawaiite composition between the Pahute Mesa, Trail Ridge, and Gold Flat outflow sheets and above the Gold Flat Member outside of the Black Mountain caldera [*Ekren et al.*, 1971; D. C. Noble, unpublished data, 1965]. Therefore it is reasonable to speculate that both the very rapid development of highly evolved comendite magma and the geologically almost instantaneous reinception of Trail Ridge Plinean eruption (perhaps from a central vent [e.g., *Hildreth and Mahood*, 1986]) and large-volume ash-flow volcanism were the result of injection or movement of more mafic magma into the middle and(or) upper part of the Black Mountain magma body.

Acknowledgments. This study was partly supported by the Nevada Nuclear Waste Projects Office and the National Science Foundation. Field work in the Black Mountain area, conducted in 1983, would have been impossible without the support and personal commitments of time and effort of Harley W. Dickensheets, Chief, Project Management Branch, 554RG/DOX. We thank C. Sherman Grommé, Duane Champion, and Scott Bogue for constructive reviews of an earlier version of the manuscript and for calling our attention to several

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important recent publications. Some of the chemical data utilized was obtained by our colleague T. A. Vogel.

REFERENCES

Bath, G. D., Aeromagnetic anomalies related to remanent magnetism in volcanic rock, Nevada Test Site, Nevada Test Site, edited by E. B. Eckel, *Mem. Geol. Soc. Am.*, *110*, 15-146, 1968.

Bogue, S. W., and K. A. Hoffman, Morphology of geomagnetic reversals, *Rev. Geophys.*, *25*, 910-916, 1988.

Byers, F. M., W. J. Carr, P. P. Orkild, W. D. Quinlivan, and K. A. Sargent, Volcanic suites and related cauldrons of the Timber Mountain-Oasis Valley caldera complex, southern Nevada, *U.S. Geol. Surv. Prof. Pap.*, *919*, 70 pp., 1976.

Carr, W. J., Regional structural setting of Yucca Mountain, southwestern Nevada, and late Cenozoic rates of tectonic activity in part of the southwestern Great Basin, Nevada and California, *U. S. Geol. Surv. Open File Rep.*, *84-854*, 114 pp., 1984.

Clement, B. M., and D. V. Kent, Short polarity intervals within the Matuyama: Transitional field records from hydraulic piston cored sediments from the North Atlantic, *Earth Planet. Sci. Lett.*, *81*, 253-264, 1987.

Drexler, J. W., T. J. Bornhorst, and D. C. Noble, Trace-element sanidine/glass distribution coefficients for peralkaline silicic rocks and their implications to peralkaline petrogenesis, *Lithos*, *17*, 265-271, 1983.

Ekren, E. B., R. E. Anderson, C. L. Rogers, and D. C. Noble, Geology of the northern Nellis Air Force Bombing and Gunnery Range, Nye County, Nevada, *U.S. Geol. Surv. Prof. Pap.*, *651*, 91 pp., 1971.

Ellwood, B. B., Estimates of flow direction for calc-alkaline welded tuffs and paleomagnetic data reliability from anisotropy of magnetic susceptibility measurements: Central San Juan Mountains, southwest Colorado, *Earth Planet. Sci. Lett.*, 59, 303-314, 1982.

Fisher, R., Disperson on a sphere, *Proc. R. Soc. London, Ser. A*, *217*, 295-305, 1953.

Geissman, J. W., Paleomagnetism of ash-flow tuffs: Micro-analytical recognition of TRM components, *J. Geophys. Res.*, *85*, 1487-1499, 1980.

Gillett, S. L., Paleomagnetism of Tertiary tuffs from the "Lava Dam", southern House Range, Utah, *Geophys. Res. Lett.*, *9*, 1025-1028, 1982.

Gose, W. A., Paleomagnetic studies of Miocene ignimbrites from Nevada, *Geophys. J. R. Astron. Soc.*, *20*, 241-252, 1970.

Grommé, C. S., E. H. McKee, and M. C. Blake, Jr., Paleomagnetic correlations and potassium-argon dating of middle Tertiary ash-flow sheets in the eastern Great Basin, Nevada and Utah, *Geol. Soc. Am. Bull.*, *83*, 1619-1638, 1972.

Hildreth, W., and G. A. Mahood, Ring-fracture eruption of the Bishop Tuff, *Geol. Soc. Am. Bull.*, *97*, 396-403, 1986.

Hoffman, K. A., Transitional field behavior from Southern Hemisphere lavas: Evidence for two-stage reversals of the geodynamo, *Nature*, *320*, 228-232, 1986.

Hoffman, K. A., Ancient magnetic reversals: Clues to the geodynamo, *Sci. Am.*, *258*, 76-83, 1988.

Irving, E., and G. A. Irving, Apparent polar wander paths Carboniferous through Cenozoic and the assembly of Gondwana, *Geophys. Surv.*, *5*, 141-188, 1982.

Jezek, P. A., and D. C. Noble, Natural hydration and ion exchange of obsidian: An electron microprobe study, *Am. Mineral.*, *63*, 266-273, 1978.

Kistler, R. W., Potassium-argon ages of volcanic rocks in Nye and Esmeralda Counties, Nevada, Nevada Test Site, edited by E. B. Eckel, *Mem. Geol. Soc. Am.*, *110*, 251-262, 1968.

Knight, M. D., G. P. L. Walker, B. B. Ellwood, and J. F. Diehl, Stratigraphy, paleomagnetism and magnetic fabric of the Toba tuffs: Constraints on the sources and eruptive styles, *J. Geophys. Res.*, *91*, 10,355-10,382, 1986.

Laj, C., S. Guitton, C. Kissel, and A. Mazaud, Complex behavior of the geomagnetic field during three successive polarity reversals, 11-12 m.y. B.P., *J. Geophys. Res.*, *93*, 11,655-11,666, 1988.

Leat, P. T., R. Macdonald, and R. L. Smith, Geochemical evolution of the Menengai caldera volcano, Kenya, *J. Geophys. Res.*, *89*, 8571-8592, 1984.

Mahood, G. A., and R. E. Drake, K-Ar dating young rhyolitic rocks: a case study of the Sierra La Primavera, Jalisco, Mexico, *Geol. Soc. Am. Bull.*, *93*, 1232-1241, 1982.

Mankinen, E. A., M. Prévot, C. S. Grommé, and R. S. Coe, The Steens Mountain (Oregon) geomagnetic polarity transition, 1, Directional history, duration of episodes and rock magnetism, *J. Geophys. Res.*, *90*, 10,393-10,416, 1985.

McElhinny, M. W., *Paleomagnetism and Plate Tectonics*, 357 pp., Cambridge University Press, Cambridge, 1973.

McElhinny, M. W., and R. T. Merrill, Geomagnetic secular variation over the past 5 m.y., *Rev. Geophys.*, *13*, 687-708, 1975.

Merrill, R. T., and M. W. McElhinny, *The Earth's Magnetic Field- Its History, Origin and Planetary Perspective*, 420 pp., Academic, San Diego, Calif., 1983.

Nairn, A. E. M., P. D. Rowley, and J. J. Anderson, Paleomagnetism of selected Tertiary volcanic units, southern Utah, *Spec. Pap. Geol. Soc. Am.*, *160*, 63-88, 1975.

Noble, D. C., Gold Flat Member of the Thirsty Canyon Tuff- A pantellerite ash-flow sheet in southern Nevada, *U.S. Geol. Surv. Prof. Pap.*, *525-B*, B85-B90, 1965.

Noble, D. C., Stress-corrosion failure and the hydration of glassy silicic rocks, *Am. Mineral.*, *53*, 1756-1759, 1968.

Noble, D. C., Loss of sodium from crystallized comendite welded tuffs of the Miocene Grouse Canyon Member of the Belted Range Tuff, Nevada. *Geol. Soc. Am. Bull.*, *81*, 2677-2687, 1970.

Noble, D. C., and R. L. Christiansen, Black Mountain volcanic center, Guidebook to the Geology of Four Tertiary Volcanic Centers in Central Nevada, *Rep. Nev. Bur. Mines Geol.*, *19*, 27-34, 1974.

Noble, D. C., and D. F. Parker, Peralkaline silicic volcanic rocks of the western United States, *Bull. Volcanol.*, *38*, 803-827, 1975.

Noble, D. C., and S. I. Weiss, Stratigraphy of the Thirsty Canyon Tuff and its bearing on the eruptive history of the Black Mountain volcanic center, southern Nevada (abstract), *Eos Trans. AGU*, *44*, 1261, 1986.

Noble, D. C., R. E Anderson, E. B. Ekren, and J. T. O'Connor, Thirsty Canyon Tuff of Nye and Esmerelda Counties, Nevada, *U.S. Geol. Surv. Prof. Pap.*, *475-D*, D24-D27, 1964.

Noble, D. C., G. S. Bath, R. L. Christiansen, and P. O. Orkild, Zonal relations and paleomagnetism of the Spearhead and Rocket Wash Members of the Thirsty Canyon Tuff, southern Nevada, *U.S. Geol. Surv. Prof. Pap.*, *600-C*, C61-C65, 1968.

Noble, D. C., W. Rigot, and H. R. Bowman, Geochemistry of rare-earth elements in silicic glasses from some highly differentiated ash-flow tuffs and lavas, *Spec. Pap. Geol. Soc. Am.*, *180*, 77-85, 1979.

Noble, D. C., T. A. Vogel, S. I. Weiss, and J. W. Erwin, Integrated field, trace-element, and paleomagnetic determination of the stratigraphic relations and source areas of ash-flow sheets of the Black Mountain and Stonewall Mountain volcanic centers, Nevada (abstract), *Eos Trans. AGU*, *64*, 336, 1983.

Noble, D. C., T. A. Vogel, S. I. Weiss, J. W. Erwin, E. H. McKee, and L. W. Younker, Stratigraphic relations and source areas of ash-flow sheets of the Black Mountain and Stonewall Mountain volcanic centers, Nevada, *J. Geophys. Res.*, *89*, 8593-8602, 1984.

Noble, D. C., E. H. McKee, and S. I. Weiss, Nature and timing of pyroclastic and hydrothermal activity and mineralization at the Stonewall Mountain volcanic center, southwestern Nevada, *Isochron West*, *51*, 25-28, 1988.

Novak, S. W., Eruptive history of the rhyolitic Kane Springs Wash volcanic center, Nevada, *J. Geophys. Res.*, *89*, 8603-8615, 1984.

Reynolds, R. L., Paleomagnetism of welded tuffs of the Yellowstone Group, *J. Geophys. Res.*, *82*, 3677-3693, 1977.

Reynolds, R. L., M. R. Hudson, and K. Hon, Paleomagnetic evidence for the timing of collapse and resurgence of the Lake City Caldera, San Juan Mountains, Colorado, *J. Geophys. Res.*, *91*, 9599-9613, 1986.

Riehle, J. R., Calculated compaction profiles of rhyolitic ash-flow tuffs, *Geol. Soc. Am. Bull.*, *84*, 2193-2216, 1973.

Rogers, C. L., E. B. Ekren, D. C. Noble, and J. E. Weir, Geologic map of the northern half of the Black Mountain quadrangle, Nye County, Nevada, *U.S. Geol. Surv. Misc. Geol. Invest. Map, I-545*, 1968.

Rosenbaum, J. G., Paleomagnetic directional dispersion produced by plastic deformation in a thick Miocene welded tuff, Southern Nevada: Implications for welding temperatures, *J. Geophys. Res.*, *91*, 12,817-12,834, 1986.

Schlinger, C. M., J. G. Rosenbaum, and D. R. Veblen, Fe-oxide microcrystals in welded tuff from southern Nevada: Origin of remanence carriers by precipitation in volcanic glass, *Geology*, *16*, 556-559, 1988.

Shuey, R. T., C. F. Caskey, and M. G. Best, Distribution and paleomagnetism of the Needles Range Formation, Utah and Nevada, *Am. J. Sci.*, *276*, 924-968, 1976.

215622022b, 1989, B5, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB094iB05p06075 by Curtin University Library, Wiley Online Library on [24/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Smith, R. L., Zones and zonal variations in welded ash flows, *U.S. Geol. Surv. Prof. Pap., 354-F*, 149-159, 1960.

Smith, R. L., Ash-flow magmatism, *Geol. Soc. Am. Spec. Pap. 180*, 5-27, 1979.

Vogel, T. A., D. C. Noble, and L. W. Younker, Chemical evolution of a high-level magma system: The Black Mountain volcanic center, southern Nevada, *Rep. UCRL-53444*, 49 pp. Lawrence Livermore Natl. Lab., Livermore, Calif., 1983.

Vogel, T. A., R. A. Ryerson, D. C. Noble, and L. W. Younker, Limits to magma mixing based on chemistry and mineralogy of pumice fragments from a chemically zoned ash-flow sheet, *J. Geol., 95*, 659-670, 1987.

Vogel, T. A., D. C. Noble and L. W. Younker, Evolution of a chemically zoned magma body: The Black Mountain volcanic center, southwestern Nevada, *J. Geophys. Res.*, this issue.

Weiss, S. I., and D. C. Noble, Paleomagnetic constraints on the rate and duration of magmatic evolution of the Black Mountain volcanic center, southern Nevada (abstract), *Eos Trans. AGU, 64*, 884, 1983.

Weiss, S. I., and D. C. Noble, Stonewall Mountain volcanic center, southern Nevada: Stratigraphic, structural, and facies relations of outflow sheets, near-vent tuffs, and intracaldera units, *J. Geophys. Res.*, this issue.

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(Received July 13, 1987;
revised December 27, 1988;
accepted December 30, 1988.)

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